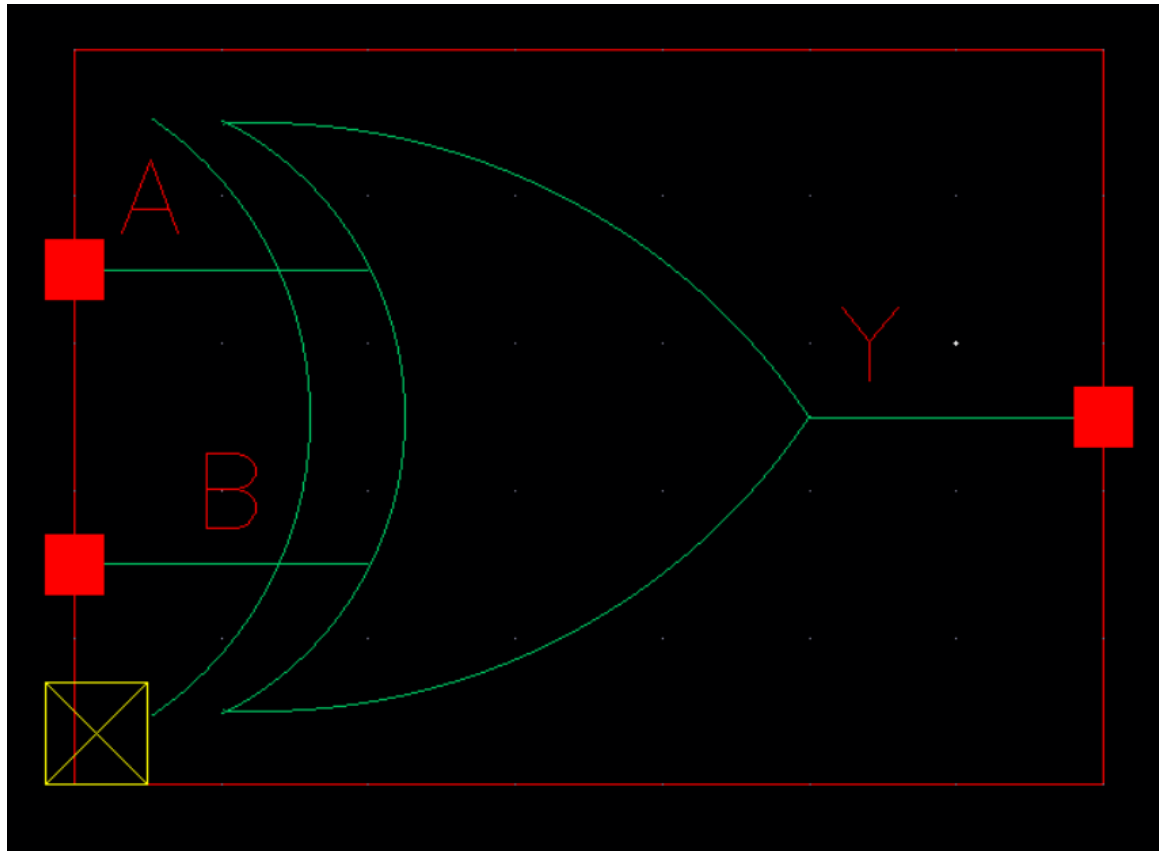
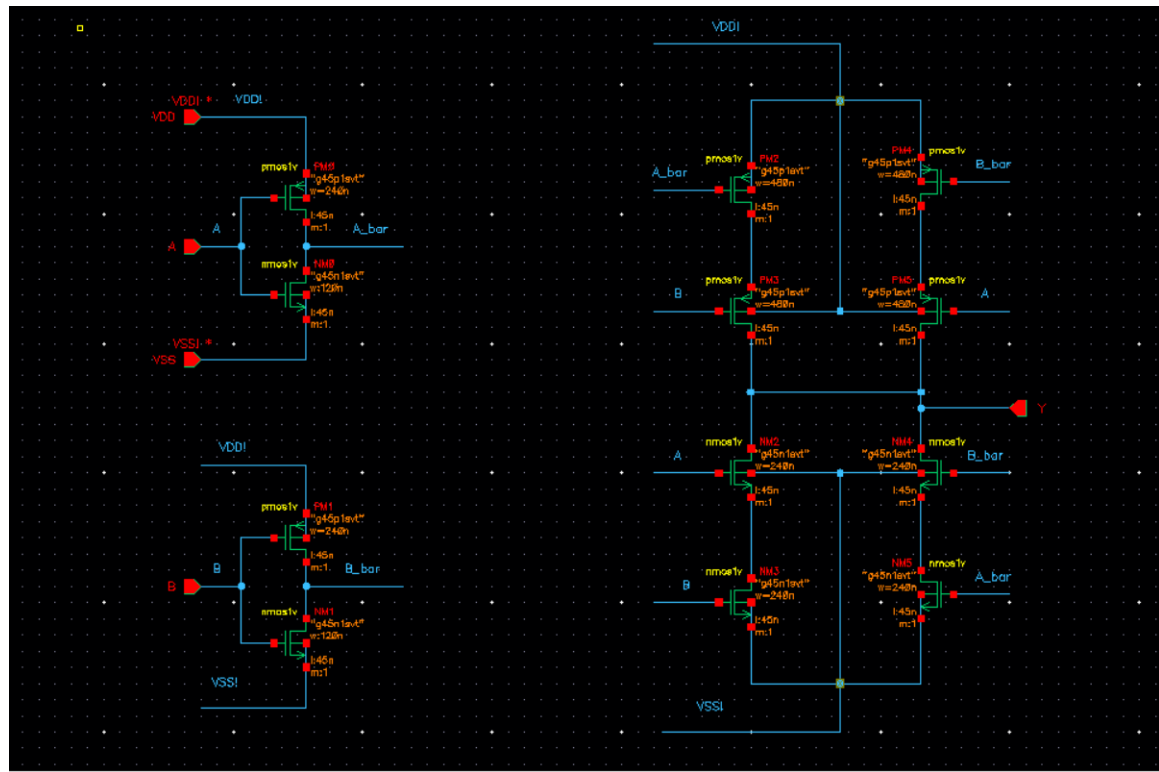
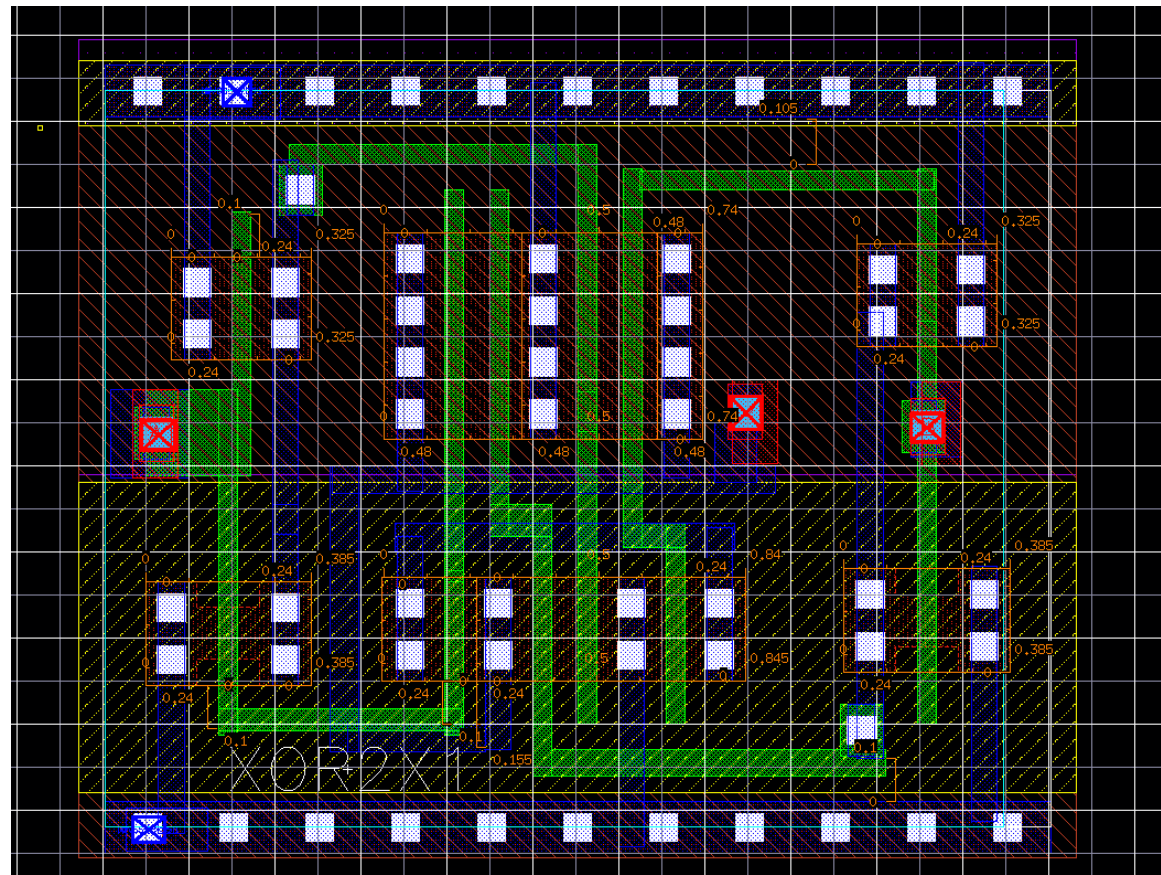


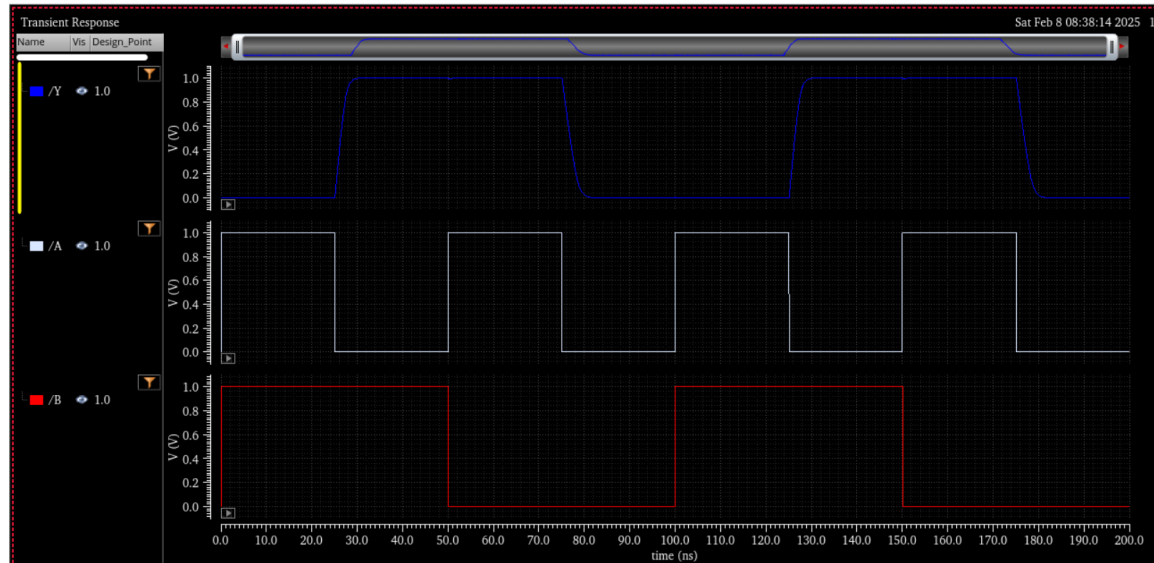
Library Name:	cgm	
Cell Name:	XOR2XI	
Function/Truth Table:		
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0
Propagation Delay (path based):		
Transition		Delay (s)
A → Y ↑ (T_PLH)		1.397n
A → Y ↓ (T_PHL)		1.679n
B → Y ↑ (T_PLH)		1.398n
B → Y ↓ (T_PHL)		1.675n
Output Rise Time (path based):		
Transition		Rise Time (s)
A → Y ↑		2.7n
B → Y ↑		2.68n
Output Fall Time (path based):		
Transition		Fall Time (s)
A → Y ↓		3.226n
B → Y ↓		3.204n

Layout Height: 1.71 μ m**Layout Width: 2.2 μ m****Layout Area: 3.762 μ m****Symbol with Port Names:**

Schematic:

Layout:

Functional Simulation Waveforms



Comments/Notes:

The XOR2X1 gate was successfully tested, confirming correct functional verification. The propagation delays remain within acceptable limits, with slight variations due to extracted parasitics. Compared to the pre-layout results, rise times have increased, while fall times have slightly improved, indicating the influence of interconnect capacitances and transistor sizing. Functional simulation verified correct logic operation, ensuring the gate meets performance requirements for integration into larger circuit designs.